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# lab-03 OR 04

## lab-03

Submitted to:-

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# Task :- 01

Assume that the following variables are defined:

```
int age;
double pay;
char section;
```

Write a single cin statement that will read input into each of these variables.

```
#include<iostream>
using namespace std;
int main(){
    int age=22;
    double pay = 45000.75;
    char section = 'B';
    cout << "Age:" << age << endl;
    cout << "Pay:" << pay << endl;
    cout << "section:" << section << endl;
    return 0;
}
```

**Output:-**

```
Age: 22
Pay: 45000.75
Section: B
```

# Task :- 02

```
#include<iostream>
  using namespace std;
int main(){
cout<<"Expression 28/4-2="<<28/4-2<<endl;
 cout<<"Expression 6+12*2-8="<<6+12*2-8<<endl;
cout<<"Expression 4+8*2="<<4+8*2<<endl;
cout<<"Expression 6+17%3-2="<<6+17%3-2<<endl;
cout<<"Expression 2+22*(9-7)="<<2+22*(9-7)<<endl;
cout<<"Expression (8+7)*2="<<(8+7)*2<<endl;
cout<<"Expression (16+7)%2-1="<<(16+7)%2-1<<endl;
cout<<"Expression 12/(10-6)="<<12/(10-6)<<endl;
cout<<"Expression (19-3)*(2+2)/4="<<(19-3)*(2+2)/4<<endl;
  return 0;
}
```

**output:**

```
Expression 28/4-2=5
Expression 6+12*2-8=22
Expression 4+8*2=20
Expression 6+17%3-2=6
Expression 2+22*(9-7)=46
Expression (8+7)*2=30
Expression (16+7)%2-1=0
Expression 12/(10-6)=3
Expression (19-3)*(2+2)/4=16
```

---

**Task :-03**

```
#include<iostream>
  using namespace std;
int main(){
    int units=10;
```

```
    float mass = 2.5f;
    double weight;
    weight = mass * units; //automatic conversion
cout << " weight is: " << weight << endl;
    cout << "Correct Option is B" << endl;
    return 0;
}
```

**Output:-**

Weight is:25
correct option is B

---

## Task :-04

```
#include<iostream>
    using namespace std;
int main (){
        int a, b=2;
        float c=4.2f;
    a = b*c; // 8.4->8
 cout <<"b* c = " << b*c << endl;
cout << " value stored in a = " << a << endl;
return 0;
}
```

**Output:-**

b*c = 8.4
value stored in a = 8

---

## Task:-05

```
#include<iostream>
 using namespace std;
int main(){
    int qty = 10;
    int salesReps = 4;
    double unitsEach;
```

```
cout<< "qty =" <<qty<< endl;
cout << "salesReps=" <<salesReps<< endl;
cout<< "UnitsEach(with type cast)=" << UnitsEach<<endl;
return 0;
}.
```

**Output:-** qty = 10
salesReps = 4
UnitsEach(with type cast) = 2.5

---

## Task :-06

```
#include<iostream>
(#include<cstd)^
using namespace std;
int main(){

srand (time(0));
int num1= rand()% 900+100;
int num2= rand()% 900+100;
int answer =num1+num2;
cout <<" "<<num1<<endl;
cout <<"+"<<num 2<< endl;
cout <<" --------"<< endl;
cout<< "\n Press Enter to see the answer...";
cin.get ();
cout <<"\n" <<num1<< endl;
cout << "+" << num2<<endl;
cout <<" --------"<< endl;
cout << " " << answer<<endl;
return0;

}
```

**Output:-**

```
 846
+150
------
```

---

```cpp
#include <iostream>
using namespace std;
int main() {
    int num = 5;
    if (num >= 0)
        cout << "Positive";
    else
        cout << "Negative";
    return 0;
}
```

Positive
[Program finished]

Expression 28 / 4 - 2 = 5
Expression 6 + 12 * 2 - 8 = 22
Expression 4 + 8 * 2 = 20
Expression 6 + 17 % 3 - 2 = 6
Expression 2 + 22 * (9 - 7) = 46
Expression (8 + 7) * 2 = 30
Expression (16 + 7) % 2 - 1 = 0
Expression 12 / (10 - 6) = 3
Expression (19 - 3) * (2 + 2) / 4 = 16

[Program finished]

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Expression 28 / 4 - 2 = " << 28 / 4 - 2 << endl;
    cout << "Expression 6 + 12 * 2 - 8 = " << 6 + 12 * 2 - 8 << endl;
    cout << "Expression 4 + 8 * 2 = " << 4 + 8 * 2 << endl;
    cout << "Expression 6 + 17 % 3 - 2 = " << 6 + 17 % 3 - 2 << endl;
    cout << "Expression 2 + 22 * (9 - 7) = " << 2 + 22 * (9 - 7) << endl;
    cout << "Expression (8 + 7) * 2 = " << (8 + 7) * 2 << endl;
    cout << "Expression (16 + 7) % 2 - 1 = " << (16 + 7) % 2 - 1 << endl;
    cout << "Expression 12 / (10 - 6) = " << 12 / (10 - 6) << endl;
    cout << "Expression (19 - 3) * (2 + 2) / 4 = " << (19 - 3) * (2 + 2) / 4 << endl;

    return 0;
}
```

```cpp
#include <iostream>
using namespace std;

int main() {
    int units = 10;
    float mass = 2.5f;
    double weight;

    weight = mass * units; // automatic conversion

    cout << "Weight is: " << weight << endl;
    cout << "Correct Option is B" << endl;

    return 0;
}
```

```
Weight is: 25
Correct Option is B

[Program finished]
```

# Home Task

## Task :- 01

1:- **Errors:-**

```
void main() →
Cin << number1;
 cin << number2;
number1 + number2 = sum;
cout "The Sum is" sum;
#include<iostream>
```

2:- **Errors:-**

```
#include<iostream
 using namespace std
  void main()
cin >>no.1, no.2;
quotient= float<static-cast
   (number1)/number 2;
cout<< quotient
```

---

## Task :- 02

```
#include<iostream>
using> namespace std;
int main(){
int v1 = 28, v2 = 32, v3 = 37, v4 = 24,
v5 = 33;
int sum;
 double average;
sum = v1 + v2 + v3 + v4 + v5;
cout << " sum=" << sum << endl;
cout << " Average = " << average << endl;
return 0;
}
```

**Output:-**

Sum = 154

Average = 30.8

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b = 2;
    float c = 4.2f;

    a = b * c; // 8.4 -> 8

    cout << "b * c = " << b * c << endl;
    cout << "Value stored in a = " << a << endl;

    return 0;
}
```

```
b * c = 8.4
Value stored in a = 8

[Program finished]

qty = 10
salesReps = 4
unitsEach (with type cast) = 2.5

[Program finished]
```

```cpp
#include <iostream>
using namespace std;

int main() {
    int qty = 10;
    int salesReps = 4;
    double unitsEach;

    // Without cast -> integer division -> 2
    // With cast -> float division -> 2.5

    unitsEach = static_cast<double>(qty) / salesReps;

    cout << "qty = " << qty << endl;
    cout << "salesReps = " << salesReps << endl;

    cout << "unitsEach (with type cast) = " << unitsEach << endl;

    // 2 more ways to write same thing
    // unitsEach = (double)qty / salesReps;
    // unitsEach = double(qty) / salesReps;

    return 0;
}
```

```cpp
#include <iostream>
#include <cstdlib> // for rand() and srand()
#include <ctime>   // for time()
using namespace std;

int main() {
    // Seed for random numbers
    srand(time(0));

    int num1 = rand() % 900 + 100; // 100 to 999 random number
    int num2 = rand() % 900 + 100;
    int answer = num1 + num2;

    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "---------" << endl;

    cout << "\nPress Enter to see the answer...";
    cin.get(); // pause

    cout << "\n  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "---------" << endl;
    cout << "  " << answer << endl;

    return 0;
}
```

```
 846
+150
--------
```

Press Enter to see the answer...

```
double number1, number2, sum;
...
void main() -> // ❌ ERROR 1
Cin << number1; -> // ❌ ERROR 2
cin << number2; -> // ❌ ERROR 3
number1+number2 = sum; -> // ❌ ERROR 4
cout  "The sum is " sum; -> // ❌ ERROR 5
// #include <iostream> missing -> ❌ ERROR 6
```

```
#include <iostream -> // ❌ ERROR 1: > missing
using namespace std -> // ❌ ERROR 2: ; missing
void main() -> // ❌ ERROR 3
cin >> number1, number2; -> // ❌ ERROR 4
quotient = float<static_cast>(number1)/number2; -
cout << quotient -> // ❌ ERROR 6: ; missing
```

```cpp
#include <iostream>
using namespace std;

int main() {
    int v1 = 28, v2 = 32, v3 = 37, v4 = 24, v5 = 33;
    int sum;
    double average;

    sum = v1 + v2 + v3 + v4 + v5;
    average = sum / 5.0; // 5.0 se divide taake decimal aye

    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;

    return 0;
}
```

```
Sum = 154
Average = 30.8

[Program finished]
```